

Multiscale Modeling of Biological Systems - Computational Neuroscience - Assignment 4

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1 Introduction

This study uses representational similarity analysis (RSA) [4] to compare the alignment of real brain data with different computational models of auditory processing. The provided brain data originate from the superior temporal gyrus (STG), a region in the auditory cortex that is believed to represent intermediate acoustic-to-semantic sound representations. The assignment provides three pretrained model families with increasing architectural complexity, and YAMNet embeddings as an additional reference model.

The three provided architectures differ in their input representation and biological inspiration. The WaveformModel operates directly on raw waveforms and contains no convolutional or recurrent structure. The UninspiredModel operates on spectrograms and uses convolutional filters over frequency and time, loosely mirroring tonotopic and spectrotemporal filtering in early auditory cortex. The InspiredModel also takes spectrograms and adds a recurrent GRU to integrate information over time, with ReLU6 providing a bounded activation. Furthermore, for this assignment we have constructed a biological model inspired by auditory-cortex anatomy.

Model	Input	Conv.	Recurrent	Activation	Parameters	Motivation
WaveformModel	raw waveform	No	No	Tanh	8,893,554	Baseline with no auditory structure assumed; MLP transformations of the raw signal.
UninspiredModel	spectrogram	Yes	No	Tanh	43,122	Convolutional filters over frequency and time loosely mirror tonotopic, spectrotemporal filtering of early auditory cortex.
InspiredModel	spectrogram	Yes	Yes (GRU)	ReLU6	203,314	Adds recurrence to integrate information over time, mirroring downstream auditory regions such as STG; ReLU6 caps activation.

In this report the question will be addressed whether increasing model complexity not only improves task performance but also makes the network more aligned with brain representations. The analysis therefore compares untrained and trained models, different architectural choices, layer depth, and a larger pretrained audio model (YAMNet).

The study is primarily *contrastive*: we vary one factor at a time (training, architecture, depth) and test how each changes the alignment between model and brain representations. It also has a *normative* element, because all models were trained only on a sound-classification task and never on brain data; alignment with STG therefore tests whether the task of recognizing sounds produces brain-like representations. The design of our own biologically inspired model in Part II adds a *constructive* component, in which anatomical knowledge is built into the architecture.

To address the constructive component of this assignment, we designed a biological model that explicitly mirrors the anatomical hierarchy of the human auditory cortex. Because there are many different brain regions involved in sound processing, it was beyond the scope of this assignment to incorporate all of them. Instead, we identified a core pathway of the most important regions using the multimodal human cortical parcellation atlas (HCP-MMP1.0) [10], and mapped the hierarchical flow between them using human effective connectivity literature [11].

Our model incorporates an input gateway at the primary auditory cortex (A1), which receives subcortical input (where sound is decomposed into frequencies) and processes basic, localized spectrotemporal features. From A1, the signal splits into parallel processing streams in the lateral and medial belt (LBelt and MBelt) to process intermediate auditory features. These two parallel streams then converge and are integrated in the parabelt (PBelt). From there, the signal projects to the higher-order association area A4, and finally to the superior temporal sulcus (A5/STS), which integrates long-term temporal context to form semantic meaning.

Constructing this architecture presented several practical challenges. First, without in-depth knowledge of the anatomy of the brain, it was difficult to decide exactly which regions to include or exclude. Furthermore, as was also discussed during the lecture, there is no universally agreed-upon map of the human auditory cortex, and different methodologies yield diverging connectomes. For example, structural scans like Diffusion Tensor Imaging reveal physical pathways but cannot reliably determine the direction in which information is flowing.

To overcome these limitations, we simplified the complex biological graph into a computationally feasible feed-forward hierarchy ($A1 \rightarrow \text{LBelt/MBelt} \rightarrow \text{PBelt} \rightarrow A4 \rightarrow A5/\text{STS}$). Finally, we assigned network operations that preserve specific biological inductive biases: convolutions to preserve tonotopy, parallel branches for belt routing, and recurrence for temporal integration.

2 Methods

2.1 Data and model activations

The analysis used the 288 natural sounds of the Santoro dataset [1], divided into six stimulus categories of 48 sounds each: speech, voice (non-speech vocalizations), animal, music, nature and tools. The brain data consist of fMRI response estimates (beta weights) of a single participant in 5521 STG voxels, giving a 288×5521 matrix with one response pattern per sound. The betas were provided in two separate sets of 216 and 72 different sounds, so each sound was measured once and the reliability of the brain RDM (a noise ceiling) could not be estimated. The hemisphere of the STG voxels was not specified in the dataset. YAMNet, a MobileNet-based network [3] trained on AudioSet [2], was provided as activations for 15 layers, from layer01relu through layer14relu and the final embedding.

For each of the four custom model types—waveform, uninspired, inspired, and biological—five trained runs were analyzed. For every trained run, an independently initialized untrained model of the same architecture was also evaluated. Activations were extracted for every layer while preserving the same sound ordering as the brain data. The custom models were trained on the ESC-50 dataset [5]; none of the models was trained on brain data, and the STG responses were used only for evaluation.

2.2 Architecture and construction of the biological model for Part II of this assignment

The biological model maps the identified anatomical regions to a specific Convolutional Recurrent Neural Network (CRNN) architecture. The input mel-spectrogram is first processed by a convolutional layer representing A1 for basic spectrotemporal feature extraction. The signal then splits into two parallel convolutional branches representing the LBelt and MBelt. These branches process intermediate auditory features in parallel, are subsequently concatenated along the channel dimension, and are fed into a convolutional layer representing the PBelt integration zone. A final convolutional layer represents A4, further expanding the feature representation. To transition from spatial to temporal processing, the remaining frequency (tonotopic) bins are averaged out. The

preserved temporal sequence is then processed by a bidirectional Gated Recurrent Unit (GRU) simulating the superior temporal sulcus (STS/A5), before the concatenated final hidden states are passed to a linear classifier.

To maximize biological plausibility, several specific inductive biases were integrated into this architecture:

First, ReLU6 was utilized as the activation function used across the network to enforce the biological constraint of a bounded firing rate, establishing a strict lower bound of zero (preventing negative neuronal firing) and an absolute maximum activation ceiling.

Second, standard batch normalization was replaced with Local Response Normalization (`nn.LocalResponseNorm`) simulating biological lateral inhibition where strongly firing frequency channels dampen adjacent neighbors [12]. This concept was discussed during the lecture and by implementing this we hope that our model is more biologically plausible. The scaling hyperparameters ($\alpha = 10^{-4}$, $\beta = 0.75$, $k = 2.0$) were adopted from established empirical derivations, originally for AlexNet [17]. The neighborhood `size` parameter was carefully scaled to match layer capacity: a localized window of 3 was used for the narrower 32-channel early layers (A1, LBelt, MBelt), while a wider window of 5 was applied in the broader PBelt and A4 layers to reflect the increased complexity and broader integration of feature maps at those stages.

Third, to model the broadening of spectral bandwidths without sacrificing the temporal fidelity required by downstream recurrent modules, asymmetric pooling was applied [13, 14, 15]. An initial symmetric pool (`kernel_size=(2, 2)`) in A1 downsamples both frequency and time to reduce baseline dimensionality and simulate initial localized pooling. All subsequent secondary regions (LBelt, MBelt, PBelt, and A4) use frequency-only asymmetric pooling (`kernel_size=(2, 1)`). This progressively compresses the tonotopic dimension while explicitly preserving the temporal sequence length for the downstream GRU.

Fourth, layer widths were determined using the principle of Capacity Scaling. This concept was also explained during the lecture. Rather than choosing arbitrary capacities, the network scales from 32 channels in A1, LBelt, and MBelt, to 64 channels in the PBelt and A4, to 128 channels in A5/STS. This progression mathematically approximates the increasing cortical volume and neuronal resources of ascending auditory regions. This 128-dimensional capacity is maintained in the GRU to balance representational richness with computational efficiency.

Fifth, the A5/STS recurrent stage utilizes a two-layer bidirectional GRU. Stacking two layers allows the model to capture more complex temporal dependencies in the auditory signal, while the bidirectional processing allows the network to integrate semantic context from both past and future acoustic cues.

Finally, to robustly evaluate the effect of training on representational alignment, five independent instances (runs) of the biological model were initialized with different random seeds. Each run was trained for exactly 10 epochs using hyperparameters identical to those of the baseline models. By using this fixed training budget, we tried to prevent overfitting on the small ESC-50 dataset. Furthermore, we hope this ensures that any observed differences in STG alignment stem purely from the model’s architectural inductive biases rather than unequal training exposure.

2.3 Representational dissimilarity matrices

For each representation, a representational dissimilarity matrix (RDM) was constructed. Two dissimilarity measures were evaluated: Euclidean distance and $1 - \text{Pearson correlation}$. Spatial and feature dimensions were flattened for each sound before computing pairwise dissimilarities. The Euclidean RDM was computed with vectorized pairwise distances, while the $1 - \text{correlation}$ RDM was obtained from the correlation matrix. The brain RDM, custom-model RDMs, and YAMNet RDMs were then compared by correlating the upper-triangular entries. Pearson and Spearman

correlations were compared as possible RDM-comparison metrics using YAMNet as a baseline.

RDM dissimilarity	RDM comparison	
	Pearson	Spearman
Euclidean	0.1151	0.1218
1 – correlation	0.0766	0.0980

The table above shows the peak YAMNet–STG alignment for all four combinations of dissimilarity measure and RDM comparison. Spearman scoring gave higher alignment than Pearson for both dissimilarity measures. This is consistent with a monotonic but nonlinear relationship between model and brain dissimilarities, so Spearman was used throughout. Euclidean RDMs gave higher alignment than 1 – correlation RDMs. However, Euclidean distance is sensitive to each sound’s overall response magnitude, so this advantage may partly reflect sound level rather than shared representational structure.

All subsequent analyses therefore use 1 – correlation RDMs, which compare the pattern of responses, and compare these RDMs using Spearman’s rank correlation. Because the same metric is applied to all models, this choice does not bias the comparisons between them. The dissimilarity measure was chosen on theoretical grounds: selecting the highest-scoring metric on the same brain data would bias the results; the table is therefore reported as a sensitivity analysis.

2.4 Statistical analysis

Five trained and five untrained runs were available per custom architecture. Mean alignment values were reported together with 95% confidence intervals (t-distribution over runs). Differences between trained and untrained runs are reported with Welch confidence intervals because we were hesitant to make assumptions about equality of variances. To assess whether alignment exceeded chance, the brain RDM was permuted 1000 times by jointly permuting its rows and columns, and the resulting null distributions were compared with the observed model–brain alignment. Max-statistic correction was applied across layers within each model and condition.

Training effects were tested by permuting trained/untrained run labels. These tests were two-sided and corrected for multiple comparisons using Bonferroni-Holm correction across layers. Architecture comparisons were similarly based on permutation of architecture labels across runs, with Holm correction across pairwise comparisons.

Seed consistency was assessed by correlating RDMs across different runs of the same architecture. Depth effects were quantified by fitting a slope to alignment as a function of relative layer depth, with a two-sided permutation test against the brain null and Holm correction across the six model-by-condition profiles.

For the waveform model, which showed negative alignment, a two-sided version of the brain-permutation test was run as an exploratory follow-up. To test whether low-level acoustics drive alignment, a control RDM was built from each sound’s level, $D_{ij} = |\log \text{RMS}_i - \log \text{RMS}_j|$, and compared with the brain and model RDMs.

As was asked for in the assignment, we focus on comparisons between models rather than on isolated R^2 -like alignment values, because the brain data come from a single participant and the models are relatively small.

3 Results

3.1 Overall model-to-brain alignment and training

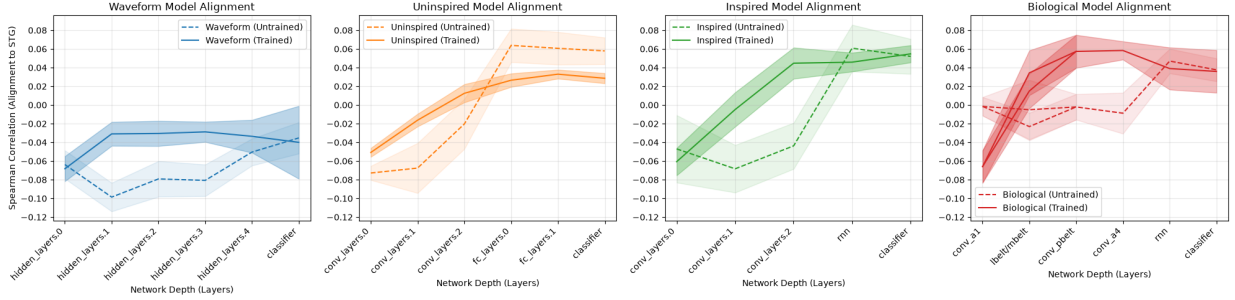


Figure 1: Spearman alignment between model-layer RDMs and the STG RDM across relative model depth. Lines distinguish trained and untrained runs; shaded regions indicate 95% confidence intervals. YAMNet is shown as a pretrained reference.

The table below summarizes the effect of training per layer. All alignment values are small, compared with 0.10 for the much larger YAMNet.

Table 2: Effect of training on alignment with STG per layer. Values are means over five runs with 95% confidence intervals. Trained and Untrained: Spearman correlation (ρ) between the model-layer RDM and the STG RDM. Δ : difference between trained and untrained alignment (trained – untrained). Seed cons.: seed consistency, the mean Spearman correlation between the RDMs of different runs of the same model and layer, for trained (T) and untrained (U) runs. Depths 1a and 1b of the biological model are its two parallel belt branches (LBelt and MBelt).

Model	Depth	Spearman ρ with STG [95% CI]			Seed cons.		
		Trained	Untrained	Δ	T	U	
waveform	0	-0.07 [-0.08,-0.05]	-0.06 [-0.08,-0.05]	-0.00 [-0.03,0.02]	0.22	0.52	
	1	-0.03 [-0.04,-0.02]	-0.10 [-0.11,-0.08]	0.07 [0.04,0.09]	0.05	0.60	
	2	-0.03 [-0.04,-0.02]	-0.08 [-0.10,-0.06]	0.05 [0.02,0.08]	0.05	0.50	
	3	-0.03 [-0.04,-0.02]	-0.08 [-0.10,-0.06]	0.05 [0.03,0.08]	0.04	0.40	
	4	-0.03 [-0.05,-0.02]	-0.05 [-0.06,-0.04]	0.02 [-0.01,0.04]	0.06	0.27	
	5	-0.04 [-0.08,-0.00]	-0.04 [-0.05,-0.02]	-0.00 [-0.06,0.05]	0.10	0.21	
uninspired	0	-0.05 [-0.06,-0.05]	-0.07 [-0.08,-0.07]	0.02 [0.01,0.03]	0.96	0.89	
	1	-0.02 [-0.02,-0.01]	-0.07 [-0.09,-0.04]	0.05 [0.01,0.09]	0.91	0.82	
	2	0.01 [0.00,0.02]	-0.02 [-0.05,0.01]	0.03 [-0.00,0.07]	0.88	0.79	
	3	0.03 [0.02,0.03]	0.06 [0.05,0.08]	-0.04 [-0.06,-0.01]	0.92	0.71	
	4	0.03 [0.03,0.04]	0.06 [0.04,0.08]	-0.03 [-0.05,-0.00]	0.90	0.69	
	5	0.03 [0.02,0.03]	0.06 [0.04,0.07]	-0.03 [-0.05,-0.01]	0.92	0.69	
inspired	0	-0.06 [-0.08,-0.05]	-0.05 [-0.08,-0.01]	-0.01 [-0.06,0.04]	0.92	0.85	
	1	-0.00 [-0.02,0.01]	-0.07 [-0.09,-0.04]	0.06 [0.03,0.10]	0.76	0.72	
	2	0.04 [0.03,0.06]	-0.04 [-0.07,-0.02]	0.09 [0.05,0.12]	0.81	0.65	
	3	0.05 [0.04,0.06]	0.06 [0.04,0.09]	-0.01 [-0.05,0.02]	0.86	0.68	
	4	0.05 [0.05,0.06]	0.05 [0.03,0.07]	0.00 [-0.02,0.03]	0.86	0.67	
	5	0.05 [0.05,0.06]	0.05 [0.03,0.07]	0.00 [-0.02,0.03]	0.86	0.67	
biological	0	-0.07 [-0.08,-0.05]	-0.00 [-0.01,0.01]	-0.06 [-0.09,-0.04]	0.92	0.93	
	1a	0.03 [0.01,0.06]	-0.01 [-0.04,0.03]	0.04 [-0.01,0.09]	0.72	0.77	
	1b	0.01 [-0.00,0.03]	-0.02 [-0.04,-0.01]	0.04 [0.01,0.07]	0.72	0.81	
	2	0.06 [0.04,0.07]	-0.00 [-0.02,0.01]	0.06 [0.03,0.09]	0.82	0.82	
	3	0.06 [0.05,0.07]	-0.01 [-0.03,0.01]	0.07 [0.04,0.10]	0.89	0.75	
	4	0.04 [0.02,0.06]	0.05 [0.03,0.06]	-0.01 [-0.04,0.02]	0.80	0.87	
	5	0.04 [0.01,0.06]	0.04 [0.03,0.05]	-0.00 [-0.03,0.03]	0.80	0.77	
	5	0.04 [0.01,0.06]	0.04 [0.03,0.05]	-0.00 [-0.03,0.03]	0.80	0.77	

In early and middle layers, training increased alignment: in the uninspired model by 0.02–0.05

(layers 0–2), in the inspired model by 0.06–0.09 (layers 1–2, where alignment turned from negative to positive) and in the biological model by 0.04–0.07 (layers 1–3).

In the deep layers the effect reversed: untrained networks aligned better than trained ones in the last three layers of the uninspired model, while the inspired model showed no clear difference in its last two layers (CIs including zero). For the biological model, in layers 1–3 the trained network showed better alignment, then in layer 4 the untrained network showed better alignment, and in the final layer they showed no difference.

As a result, the peak alignment of untrained networks was equal to or higher than that of trained networks for the provided models (per-run peak, uninspired 0.066 vs 0.033, inspired 0.062 vs 0.058). For the biological model, the trained peak was higher than the untrained peak (0.062 vs 0.048). The waveform model was negative in all layers. Training reduced its anti-alignment in the middle layers ($\Delta = 0.05$ – 0.07), but did not bring it above zero.

Seed consistency shows how training shapes the representations themselves. For the uninspired, inspired and biological models, different runs produced very similar RDMs, both untrained (0.65–0.93) and trained (0.72–0.96). For the uninspired and inspired models, training made the runs converge further, so the architecture largely determines the representational geometry and training refines it consistently. The biological model did not show this convergence: in several layers (0, 1a, 1b, 4), trained runs were as consistent as or slightly less consistent than untrained runs. The waveform model showed the opposite: untrained runs were moderately consistent (0.21–0.60), but after training, runs agreed almost not at all beyond the first layer (0.04–0.10). Each trained waveform network thus learned its own geometry, so its alignment scores describe a family of different representations rather than one stable solution.

3.2 Statistical tests of the training effect

To make a conclusion on the effect of training on brain alignment, a permutation test was used. The one-sided tests evaluated whether model–brain alignment exceeded chance; p-values were corrected using a max-statistic across layers.

Table 3: Permutation tests per layer. ρ : mean Spearman correlation between the model-layer RDM and the STG RDM over five runs. p_{adj} for Trained and Untrained: one-sided test of alignment above chance (1000 permutations of the brain data), adjusted with the max-statistic method across the layers of each model. Δ : trained – untrained alignment, with p_{adj} from a two-sided permutation test of the trained/untrained run labels, Holm-adjusted across layers. * $p_{\text{adj}} < 0.05$; $^{\dagger} p_{\text{adj}} < 0.10$. With five runs per condition, the smallest attainable p_{adj} for Δ is 0.040 (five layers), 0.048 (six layers) or 0.056 (seven layers).

Model	Depth	Trained		Untrained		Training effect	
		ρ	p_{adj}	ρ	p_{adj}	Δ	p_{adj}
waveform	0	−0.068	1.000	−0.064	0.994	−0.004	1.000
	1	−0.031	1.000	−0.099	1.000	0.068	0.048*
	2	−0.030	1.000	−0.079	0.999	0.049	0.048*
	3	−0.029	1.000	−0.081	0.999	0.052	0.048*
	4	−0.034	1.000	−0.050	0.986	0.017	0.571
	5	−0.040	1.000	−0.035	0.953	−0.005	1.000
uninspired	0	−0.051	1.000	−0.073	1.000	0.022	0.048*
	1	−0.016	0.979	−0.068	1.000	0.051	0.095 †
	2	0.013	0.723	−0.020	0.966	0.033	0.095 †
	3	0.027	0.462	0.064	0.086 †	−0.037	0.079 †
	4	0.033	0.368	0.061	0.096 †	−0.028	0.095 †
	5	0.029	0.421	0.058	0.106	−0.029	0.095 †
inspired	0	−0.061	1.000	−0.047	0.999	−0.014	1.000
	1	−0.005	0.946	−0.068	1.000	0.063	0.040*

Table 3: Permutation tests per layer (continued).

Model	Depth	Trained		Untrained		Training effect	
		ρ	p_{adj}	ρ	p_{adj}	Δ	p_{adj}
biological	2	0.045	0.211	-0.044	0.997	0.088	0.040*
	3	0.046	0.194	0.061	0.082 [†]	-0.015	0.857
	4	0.055	0.124	0.052	0.131	0.003	1.000
	0	-0.066	1.000	-0.001	0.812	-0.064	0.056 [†]
	1a	0.034	0.349	-0.005	0.843	0.039	0.286
	1b	0.015	0.726	-0.023	0.956	0.038	0.095 [†]
	2	0.057	0.106	-0.002	0.817	0.059	0.056 [†]
	3	0.058	0.101	-0.009	0.882	0.067	0.056 [†]
	4	0.039	0.284	0.047	0.189	-0.008	1.000
	5	0.036	0.326	0.038	0.269	-0.002	1.000

No layer of any model, trained or untrained, was aligned with STG above chance after correction (all $p \geq 0.082$). Training changed alignment significantly in waveform layers 1–3, uninspired layer 0 and inspired layers 1–2. In the biological model, layers 0, 2 and 3 reached $p = 0.056$, which is the smallest value attainable for its seven layers, so a significant result was impossible there by construction.

We do not consider these changes a meaningful improvement. Since no model aligned with STG above chance, training shifts alignment only within the range expected by chance. The differences are small in absolute terms ($|\Delta\rho| \leq 0.09$) and change sign with depth: training increased alignment in early and middle layers but decreased it in deep layers. All significant p-values sit exactly at the floor of the test (0.048 for six layers, 0.040 for five layers). This indicates complete separation of trained and untrained runs, but with five runs per condition it gives no information on how robust this separation is. We therefore interpret these results as weak evidence at the resolution limit of the test, not as a reliable training benefit.

3.3 Waveform model and sound-level control

The waveform model showed negative alignment in all layers, most strongly in the earliest layers (trained layer 0: $\rho = -0.068$; untrained layer 1: $\rho = -0.099$). The one-sided permutation test used so far only asks whether a model aligns with STG better than chance, so for these negative values it returns $p \approx 1$. It cannot tell whether the model is simply unrelated to STG or systematically anti-aligned, representing as similar the sound pairs that STG represents as different. A two-sided version of the same test was therefore used.

Table 4: Two-sided test of the waveform model’s alignment with STG. ρ : mean Spearman correlation between the model-layer RDM and the STG RDM over five runs. p_{adj} : two-sided brain permutation test (1000 permutations), max-statistic adjusted across layers, testing whether alignment differs from zero in either direction. * $p_{\text{adj}} < 0.05$; ** $p_{\text{adj}} < 0.01$.

Depth	Layer	Trained		Untrained	
		ρ	p_{adj}	ρ	p_{adj}
0	hidden_layers.0	-0.068	0.004**	-0.064	0.108
1	hidden_layers.1	-0.031	0.252	-0.099	0.010**
2	hidden_layers.2	-0.030	0.262	-0.079	0.037*
3	hidden_layers.3	-0.029	0.298	-0.081	0.031*
4	hidden_layers.4	-0.034	0.210	-0.050	0.207
5	classifier	-0.040	0.111	-0.035	0.428

The anti-alignment is significant in the first trained layer ($\rho = -0.068$, $p = 0.004$) and in untrained

layers 1–3 ($\rho = -0.079$ to -0.099 , $p = 0.010$ – 0.037). All other layers are negative but within the range expected by chance. The effect is therefore concentrated in the early stages, where the representation is closest to the raw audio signal, and is strongest in the untrained network. Training weakens the anti-alignment in layers 1–3 (from about -0.08 to about -0.03), but it persists in the first layer.

Because the effect sits in the layers nearest the raw waveform, we suspected a low-level acoustic property rather than learned representational content, and compared the sound-level control RDM with the brain RDM and all model layers.

Model	Depth	Trained vs loudness	Untrained vs loudness
waveform	0	-0.302	-0.552
waveform	1	-0.143	-0.671
waveform	2	-0.133	-0.602
waveform	3	-0.123	-0.525
waveform	4	-0.090	-0.422
waveform	5	-0.067	-0.368
uninspired	0	-0.184	-0.330
uninspired	1	0.012	-0.297
uninspired	2	0.071	-0.115
uninspired	3	0.086	0.198
uninspired	4	0.081	0.187
uninspired	5	0.056	0.185
inspired	0	-0.110	-0.221
inspired	1	0.080	-0.290
inspired	2	0.200	-0.240
inspired	3	0.120	0.181
inspired	4	0.099	0.162
biological	0	-0.095	-0.060
biological	1a	0.216	-0.100
biological	1b	0.127	-0.151
biological	2	0.255	-0.106
biological	3	0.292	-0.095
biological	4	0.108	0.073
biological	5	0.088	0.062

The sound-level RDM correlates positively with the STG RDM ($\rho = 0.113$), which is comparable to YAMNet’s peak alignment (0.10) and higher than any of the custom models. This result needs to be read with the stimulus set in mind: the sounds are largely level-normalized (from the 25th percentile upwards, log RMS varies by less than 0.05), and only a small group of sounds has a clearly lower level. Short sounds padded with silence do not explain this group (Spearman $\rho = -0.08$ between log RMS and the fraction of silence). The sound-level RDM therefore mainly captures how each system represents this small set of low-level sounds, not a graded code for loudness.

The waveform model shows the opposite relationship to STG: all its layers correlate negatively with sound level, most strongly in the untrained network (up to $\rho = -0.67$) and in the earliest layers. These are the same layers that were significantly anti-aligned with STG. If STG relates positively and the waveform model negatively to the same sound-level structure, their mutual alignment is expected to be negative, so the waveform model’s anti-alignment is plausibly explained by its representation of the low-level sounds. Training weakens this anti-correlation (for example from -0.67 to -0.14 in layer 1), and brain alignment becomes less negative accordingly.

The spectrogram-based models show a different pattern. In the uninspired and inspired models, early layers correlate negatively and deep layers positively with sound level, and training shifts these correlations in a depth-dependent way: wherever training increases a layer’s sound-level correlation, its brain alignment tends to increase too. The biological model follows the same pattern: while the untrained network shows weak or negative correlations with sound level, training produces clearly positive correlations in its middle stages (up to $\rho = 0.292$ in layer 3, A4). Part of the training effects reported above may therefore reflect how the models represent the low-level sounds, rather than higher-level representational content.

3.4 Effects of architectural choices on brain alignment

To compare architectures, each run was summarized by its best layer (peak) and by its average over layers (mean). Pairwise differences were tested by permuting architecture labels across runs, using two-sided tests with Holm–Bonferroni correction [6].

Condition	Summary	Waveform	Uninspired	Inspired	Biological
trained	peak	-0.019	0.033	0.058	0.062
trained	mean	-0.039	0.006	0.016	0.025
untrained	peak	-0.035	0.066	0.062	0.048
untrained	mean	-0.068	0.004	-0.009	0.006

The waveform model showed the lowest alignment in all settings, with negative alignment even at its best layer ($\rho = -0.019$ trained, -0.035 untrained; all $p_{\text{adj}} = 0.048$, the smallest attainable value). Its negative values do not indicate an absence of shared structure: as shown in Section 3.3, the waveform model is systematically anti-aligned with STG in its early layers, which can be traced to an inverted representation of low-level sounds. In terms of resemblance to STG’s geometry, however, it ranks below all spectrogram-based models.

Untrained, the spectrogram-based models (uninspired, inspired, and biological) did not differ significantly from one another (peak 0.066, 0.062, and 0.048; all $p > 0.38$). Trained, both the inspired and biological models reached higher peak alignment than the uninspired model (0.058 and 0.062 vs 0.033; both $p = 0.048$), though they did not differ significantly from each other ($p = 0.532$).

For the inspired model, this difference arises because training reduced the peak alignment of the uninspired model more strongly (0.066 to 0.033) than that of the inspired model (0.062 to 0.058), not because training increased alignment. However, for the biological model, training actively increased peak alignment (0.048 to 0.062). Averaged over layers, the biological model also significantly outperformed the uninspired model ($p = 0.048$), whereas the inspired model’s advantage over the uninspired model was no longer significant ($p = 0.159$), suggesting that the biological model’s advantage extends over more of its depth. These comparisons reflect differences between model representations that are robust across random seeds. They do not establish that any architecture captures STG’s representational structure: no model aligned with STG above chance on its own, and the architecture test generalizes over seeds but not over sounds.

3.5 Layer depth and STG alignment

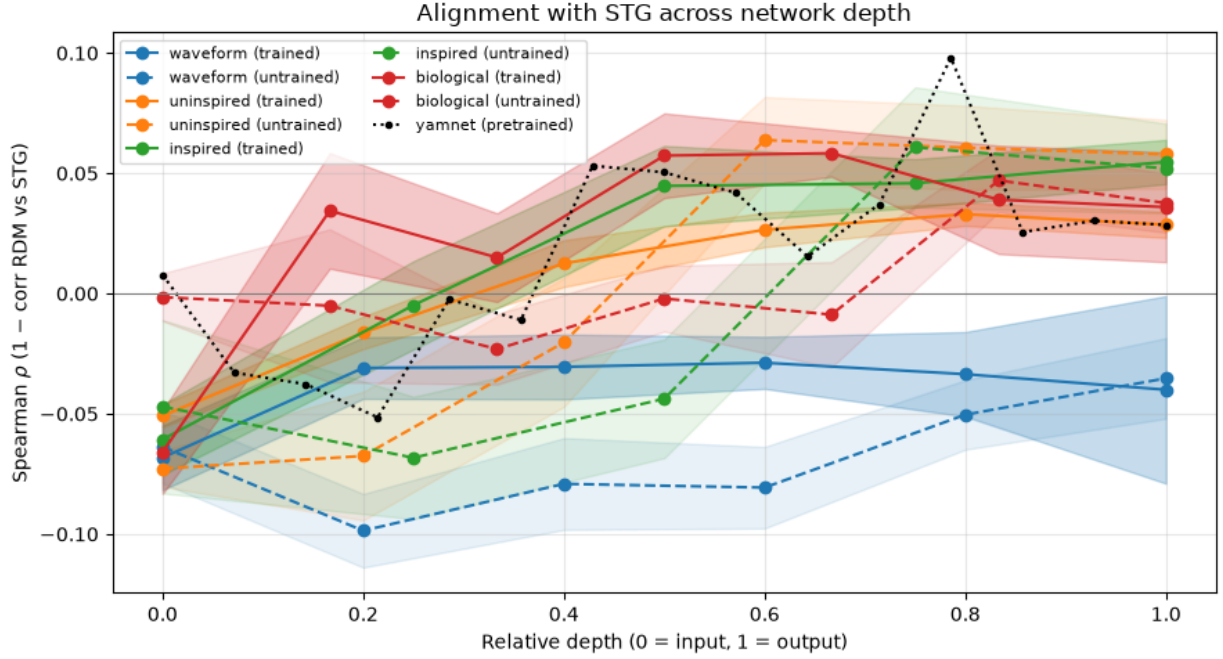


Figure 2: Alignment with STG as a function of relative model depth for trained and untrained models, with YAMNet as a reference.

Model	Trained			Untrained		
	Slope [95% CI]	p	Peak [95% CI]	Slope [95% CI]	p	Peak [95% CI]
waveform	0.019 [-0.005,0.044]	0.539	0.60 [0.25,0.95]	0.041 [0.025,0.057]	0.222	1.00 [1.00,1.00]
un-inspired	0.080 [0.073,0.086]	0.539	0.80 [0.80,0.80]	0.160 [0.137,0.183]	0.072 [†]	0.76 [0.57,0.95]
inspired	0.113 [0.096,0.129]	0.330	0.85 [0.65,1.00]	0.131 [0.095,0.167]	0.072 [†]	0.80 [0.70,0.90]
biological	0.077 [0.050,0.104]	0.539	0.60 [0.52,0.68]	0.051 [0.043,0.058]	0.539	0.87 [0.80,0.93]

Alignment with STG increased with depth in the un-inspired, inspired, and biological models. It went from negative values at the input to positive values in the deeper layers, with peaks in the later part of the network (mean relative peak depth 0.75–0.92 for the standard models, whereas the trained biological model peaked slightly earlier at intermediate depths of 0.50–0.67). The depth slopes were positive and highly consistent across runs (0.08 to 0.15 per unit of relative depth, CIs excluding zero). Tested against the brain permutation null, however, only the untrained inspired model showed a significant trend ($p = 0.036$, Holm corrected). So the increase with depth is reproducible across seeds, but mostly not larger than chance.

Training changed the shape rather than the direction of the profile: trained models rose earlier and aligned better in early and middle layers. This early rise is uniquely pronounced in the trained biological model, which spikes sharply into positive alignment at its very first parallel belt stage (relative depth ~ 0.17) before maintaining high alignment through its middle stages. Untrained models rose more steeply and reached similar or higher values in the deepest layers (for example un-inspired, slope 0.080 trained vs 0.149 untrained).

The waveform model showed no depth trend and remained negative at all depths. The peak location of its runs varied widely (CIs spanning 0.2–0.95). YAMNet, as a reference, showed a noisier profile, with its highest alignment at a late intermediate layer. Overall, the non-waveform models align best with STG in their intermediate to later layers, consistent with STG as a non-primary auditory region. The steep rise of the untrained models, however, coincides with increasing loudness structure

in their deep layers, so part of this depth effect may reflect loudness rather than a processing hierarchy.

3.6 Representational structure and category visualization

Representational structure was visualized with non-metric multidimensional scaling (MDS, unsupervised) and cross-validated linear discriminant analysis (LDA, supervised). MDS was applied directly to the 1,-,correlation RDMs used in the RSA, so each map shows the geometry that was compared with STG. Non-metric MDS preserves only the rank order of dissimilarities, matching the Spearman comparison, and, unlike t-SNE, aims to preserve all pairwise dissimilarities rather than only local neighborhoods. Kruskal stress indicates how well each two-dimensional map reflects the full RDM. Maps were aligned to the STG map with a Procrustes transformation (rotation, reflection and scaling only), so that corresponding sounds appear in comparable positions across panels.

Non-metric MDS of 1 – correlation RDMs (trained), colored by category

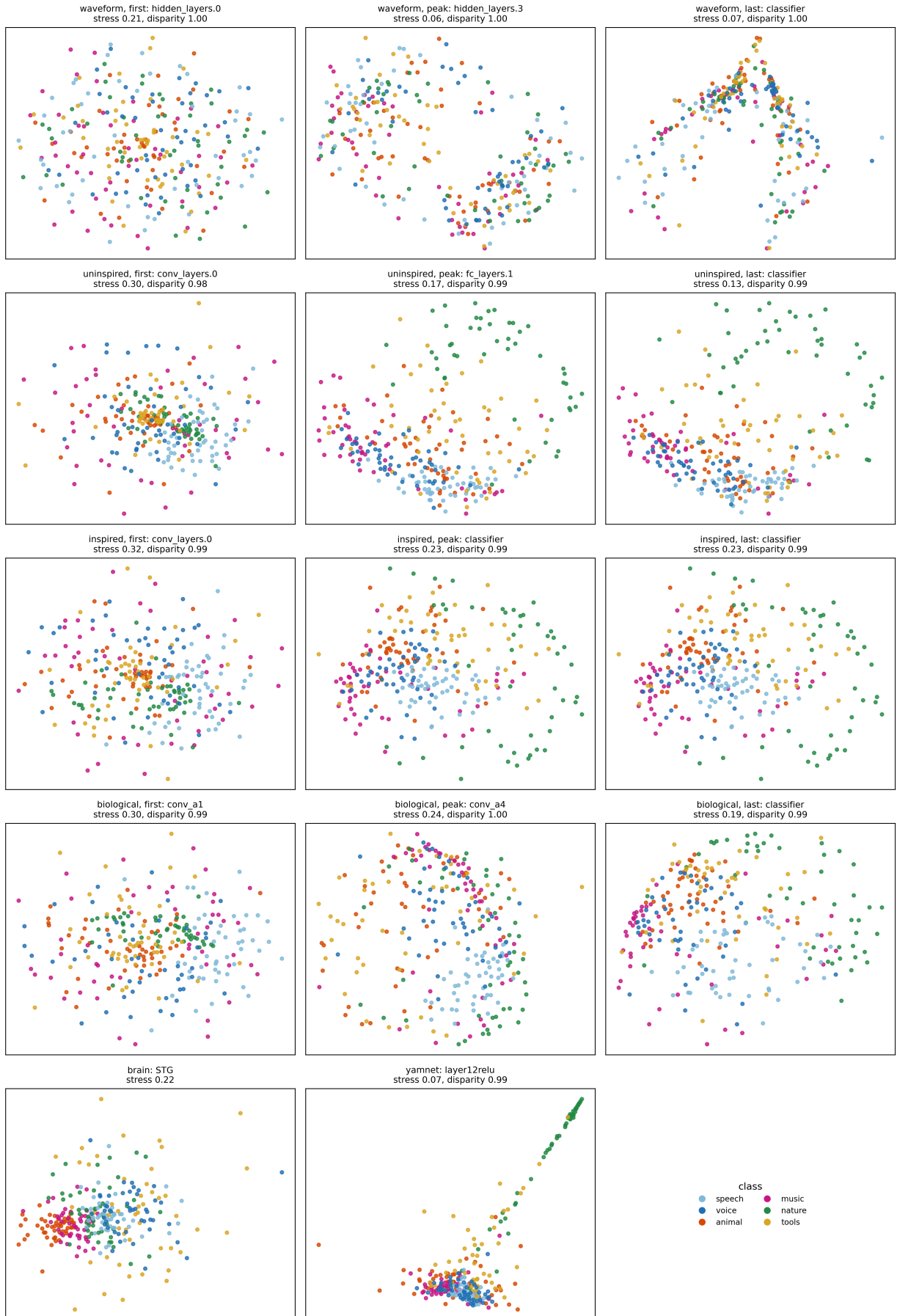


Figure 3: Non-metric MDS maps of the 1–correlation RDMs for first, peak, and last layers of representative trained runs, together with STG and YAMNet.

The figure shows non-metric MDS maps of the $1 - \text{correlation}$ RDMS for the first, peak and last layer of one representative trained run per model, alongside STG and YAMNet’s peak layer. Sounds are colored by their six categories: speech and voice (light and dark blue), animal (orange), music (purple), nature (green) and tools (yellow).

In the first layers, sounds form a diffuse cloud with only a small central cluster of tool and animal sounds. The high stress (0.21–0.32) shows that two dimensions capture this geometry poorly. In deeper layers of the uninspired and inspired models, nature sounds move to their own region, apart from the vocal and music sounds, while tools stay spread between the two. Within the vocal region, speech partly separates from voice and music, and the inspired model also groups animal sounds together. The biological model follows the same pattern, but later: at its peak layer (A4) only speech forms its own region. Nature sounds separate only in the classifier, where music and voice sit on one side, speech in the middle and nature on the other side. The waveform model forms a low-dimensional arc (stress 0.06–0.07), but the categories stay mixed along it. STG shows a different organization: animal sounds form a clear cluster on one side, overlapping with music, while speech and voice lie on the other side and tools are spread at the edges. YAMNet shows the clearest structure: nature sounds form an elongated arm separated from a compact cluster of speech, voice, music and animal sounds. Procrustes disparities of 0.98–1.00 show that no model’s map matches the STG layout, which fits the low RSA alignment. Overall, the models mainly separate nature from vocal and music sounds, while STG mainly separates animal sounds from human speech and voice.

Non-metric MDS of 1 – correlation RDMs (trained), colored by sound level



Figure 4: The same MDS representations colored by sound level (log RMS amplitude).

Coloring the same maps by sound level (log RMS) shows how much level shapes each representation. The stimuli are largely level-normalized, so the color scale was limited to the 5th–95th percentile, and the visible variation comes mainly from the quietest quarter of the sounds. In the first layer of all four models, the quietest sounds form a tight central cluster, meaning weak inputs produce similar, nondistinctive activations. In the waveform model, they stay concentrated at one point of the arc in deeper layers, which makes level one of its dominant axes. In the uninspired and inspired models, they move away from the main cloud but spread out rather than cluster. The biological model isolates them most clearly: at its peak layer (A4), they form a distinct group along the left edge of the map. In STG, level shows no systematic organization: only a few quiet sounds lie at the periphery. YAMNet’s main axis separates nature sounds from the other categories rather than by level. These patterns match the sound-level RDM correlations. The strongly negative waveform correlations reflect quiet sounds being represented as similar to all others. The weakly positive STG correlation reflects a few quiet sounds that differ from the rest. The highest positive correlation of the biological model, at A4, matches the separate group of quiet sounds in that layer.

3.7 Linear category readout

MDS shows the dominant structure of a representation, but a representation can hold category information outside its main axis of variation, for example when sound level dominates the geometry. To test whether categories are linearly readable, we used cross-validated Linear Discriminant Analysis (LDA). Sound patterns were z-scored across features, matching the 1 – correlation used in the RSA, and reduced to 30 principal components before fitting shrinkage LDA to the six category labels. Because LDA is supervised and can separate even random labels on training data, we report only held-out results: the projection of sounds left out of fitting (50/50 split) and 5-fold cross-validated balanced accuracy (chance $1/6 \approx 0.17$). MDS thus shows how a representation is organized, and LDA shows whether categories can be read out from it.

Cross-validated LDA (held-out sounds, trained models)



Figure 5: Cross-validated LDA projections and category decoding for representative layers, STG, and YAMNet.

LDA showed clear differences in how much category information is linearly available (Figure 5). The waveform model was barely above chance in all layers (0.22–0.25, chance 0.17). The uninspired and inspired models already reached 0.54 and 0.60 in their first layers and 0.63 and 0.64 in their deeper layers, so much of the category information is already present in the spectrogram input. The biological model stayed nearly flat across depth (0.55, 0.57 and 0.56 for first, peak and last layer). STG reached 0.67, higher than any of the four trained models, so its response patterns carry strong category information. YAMNet reached 0.87.

In the held-out projections of the deeper spectrogram-based layers, nature sounds separated from the vocal and music sounds, and speech separated partly from voice. In the biological model, this nature separation appeared only in the classifier. YAMNet showed the clearest structure, with speech, voice and animal sounds lying in adjacent groups and tools and nature sounds spreading along a separate arm. STG was organized differently, with animal and music sounds on one side and speech and voice on the other, consistent with its MDS map. The first uninspired layer decoded well (0.54) without showing category structure in MDS, so category information can be present without dominating the geometry.

Among the waveform, uninspired and inspired models and YAMNet, the ranking by category decoding matches the ranking by peak brain alignment. The biological model does not fit this pattern: its peak alignment is as high as that of the inspired model (0.062 vs 0.058, not significantly different), although its decoding is lower (0.57 vs 0.64). Alignment with STG therefore does not require maximal linear category separation.

4 Discussion

The results indicate that architectural structure is related to the representational geometry observed in STG, but the evidence for a reliable improvement in brain alignment through training is very limited. The waveform model remained negatively aligned with STG, whereas the spectrogram-based architectures developed positive alignment in intermediate and later layers. After training, the biological model rose earliest and reached a peak alignment as high as that of the inspired model.

The biological model. Training raised the biological model’s alignment from the parallel belt stages onwards, with the largest gains at layers 2–3 ($\Delta = 0.06$ – 0.07). Its peak lay at intermediate depth (A4) rather than near the output. It was the only model whose training increased peak alignment (0.048 to 0.062). Its peak did not differ from that of the inspired model ($p = 0.532$), but unlike the inspired model it also outperformed the uninspired model averaged over layers ($p = 0.048$), so its advantage extends over more of its depth. It reached this alignment with lower category decoding than the inspired model (0.57 vs 0.64), which fits the view that brain alignment does not follow task performance alone [7]. Architectural constraints such as parallel belt routing, lateral inhibition and asymmetric pooling are candidate explanations, but without ablations we cannot attribute the effect to any of them. Part of the alignment may also reflect sound level: A4 is the layer where the model correlates most with the sound-level RDM (0.292) and isolates the low-level sounds most clearly.

Limitations. The brain data come from a single participant, and because each sound was measured only once, no noise ceiling could be computed. The models were trained on the small ESC-50 dataset, which limits the features they can learn compared with large-scale models such as YAMNet. No layer of any model aligned with STG above chance after correction (all $p \geq 0.082$). With five runs per condition, the run-label tests have a coarse resolution (smallest corrected p-values 0.040–0.056), and for the seven-layer biological model significance was unattainable. The relative ranking of the architectures is therefore more informative than their absolute alignment.

Synthesis. Training flattened the steep late rise of the untrained models and made alignment

rise earlier, most clearly in the biological model. The waveform model’s anti-alignment is plausibly explained by its reversed representation of low-level sounds, whereas the spectrogram-based models, especially the biological model at A4, set these sounds apart as STG does. In the MDS maps, the biological model separated nature sounds from vocal and music sounds only in its classifier, and grouped the low-level sounds at the edge of its A4 map. LDA showed that it matched STG geometry as well as the inspired model while falling behind it in linear category readout. A model’s representational geometry can thus become more brain-like without maximizing linear category separability.

5 Future Work

While the biological model reached STG alignment comparable to the best provided model with parallel pathways and capacity scaling, it remains a strongly simplified model of the auditory pathway. Future work could add neuroanatomical detail and test which components drive the alignment.

Feedback. Apart from a single recurrent layer, information in our model flows only forward. The auditory system, in contrast, has extensive descending projections from higher auditory cortex to A1 and subcortical stations, which are thought to support prediction and attention. Adding feedback connections between stages, for example from the A4-like stage back to the core, would test whether top-down input brings the geometry closer to STG.

Temporal dynamics and noise. Our model processes the whole spectrogram at once, without the transmission delays between regions that shape timing in the brain. Processing the input over time with realistic inter-stage delays would add temporal dynamics that standard convolutions lack. Adding sensory and neuronal noise could further increase biological plausibility and robustness.

Spiking units. The model uses continuous rate-based activations (ReLU6), whereas biological neurons communicate with spikes. A spiking neural network could capture aspects of neural coding such as spike timing. Since fMRI measures a slow hemodynamic signal, however, its benefit may show more clearly in comparisons with electrophysiological than with fMRI data.

Training and evaluation. The models were trained with supervised labels on the small ESC-50 dataset. Larger datasets or self-supervised objectives could produce richer representations and a better fit with STG, since brain alignment does not follow classification accuracy alone [8]. On the evaluation side, data from more participants, with repeated measurements of each sound, would allow a noise ceiling to be estimated and tests to generalize across participants rather than only across random seeds. Our results also point to two direct follow-ups: ablating single components of the biological model (parallel belt routing, lateral inhibition, asymmetric pooling) to identify what drives its alignment, and controlling for sound level before interpreting alignment as shared higher-level content.

Hemispheric differences. The STG data used here did not specify the hemisphere. Auditory cortex is thought to be lateralized, with the left hemisphere attending to fast temporal changes relevant for speech and the right hemisphere favoring spectral detail relevant for pitch, music and voice [9]. With bilateral data, RSA could be run separately per hemisphere to test whether the distinction between speech, voice and non-human sounds differs between them. On the model side, two parallel streams with different temporal and spectral resolution could be compared with each hemisphere separately.

6 Conclusion

This study investigated whether increasing the architectural complexity and biological detail of auditory neural networks improves their representational alignment with the human superior tem-

poral gyrus (STG). The input representation had the clearest effect: the waveform model was anti-aligned with STG, plausibly because it represents low-level sounds in the opposite way, whereas all spectrogram-based models developed positive alignment in their intermediate and later layers. The biological model, built with parallel processing streams, asymmetric pooling and capacity scaling, was the only model whose training increased peak alignment. It reached the level of the inspired model, outperformed the uninspired model across its depth, and rose earliest, from its parallel belt stages onwards.

No model aligned with STG above chance, and with five runs per model the architecture comparisons rest on a coarse test, so these results are relative rather than absolute. Within these limits, the biological model matched the inspired model’s alignment despite lower category decoding. This suggests that optimizing for category separability alone does not make representations brain-like, and that anatomical constraints may offer a complementary route. Ablations, more participants and a control for sound level are needed to establish which parts of the architecture drive this effect.

7 References

References

- [1] Santoro, R., Moerel, M., De Martino, F., Goebel, R., Ugurbil, K., Yacoub, E., & Formisano, E. (2014). Encoding of natural sounds at multiple spectral and temporal resolutions in the human auditory cortex. *PLoS Computational Biology*, 10(1), e1003412.
- [2] Gemmeke, J. F., Ellis, D. P. W., Freedman, D., Jansen, A., Lawrence, W., Moore, R. C., Plakal, M., & Ritter, M. (2017). Audio Set: An ontology and human-labeled dataset for audio events. *Proceedings of IEEE ICASSP*, 776–780.
- [3] Howard, A. G., Zhu, M., Chen, B., Kalenichenko, D., Wang, W., Weyand, T., Andreetto, M., & Adam, H. (2017). MobileNets: Efficient convolutional neural networks for mobile vision applications. *arXiv:1704.04861*.
- [4] Kriegeskorte, N., Mur, M., & Bandettini, P. (2008). Representational similarity analysis – connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2, 4.
- [5] Piczak, K. J. (2015). ESC: Dataset for environmental sound classification. *Proceedings of the 23rd ACM International Conference on Multimedia*, 1015–1018.
- [6] Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70.
- [7] G. Tuckute, J. Feather, D. Boebinger, and J. H. McDermott. Many but not all deep neural network audio models capture brain responses and exhibit correspondence between model stages and brain regions. *PLoS Biology*, 21(12):e3002366, 2023.
- [8] M. Schrimpf, J. Kumbusi, H. Hong, N. J. Majaj, R. Rajalingham, E. B. Issa, K. Kar, P. Bashivan, J. Prescott-Roy, F. Geiger, K. Schmidt, D. L. K. Yamins, and J. J. DiCarlo. Brain-Score: Which artificial neural network for object recognition is most brain-like? *bioRxiv*, 407007, 2018.
- [9] R. J. Zatorre and P. Belin. Spectral and temporal processing in human auditory cortex. *Cerebral Cortex*, 11(10):946–953, 2001.
- [10] Glasser, M. F., Coalson, T. S., Robinson, E. C., Hacker, C. D., Harwell, J., Yacoub, E., ... & Van Essen, D. C. (2016). A multi-modal parcellation of human cerebral cortex. *Nature*, 536(7615), 171–178.

- [11] Rolls, E. T., Deco, G., Huang, C. C., & Feng, J. (2023). Human effective connectivity of the auditory, language, and semantic systems. *Brain Structure and Function*, 228(1), 227-248.
- [12] Weidler, T., Lehnen, J., Denman, Q., Sebők, D., Weiss, G., Driessens, K., & Senden, M. (2021). Biologically Inspired Semantic Lateral Connectivity for Convolutional Neural Networks. *arXiv:2105.09830*.
- [13] Hackett, T. A. (2011). Information flow in the auditory cortical network. *Hearing Research*, 271(1-2), 133-146.
- [14] Moerel, M., De Martino, F., & Formisano, E. (2014). An anatomical and functional topography of human auditory cortical areas. *Frontiers in Neuroscience*, 8, 225.
- [15] Kell, A. J., Yamins, D. L., Shook, E. N., Norman-Haignere, S. V., & McDermott, J. H. (2018). A task-optimized neural network replicates human auditory behavior, predicts brain responses, and reveals a cortical processing hierarchy. *Neuron*, 98(3), 630-644.
- [16] Esposito, M., Weidler, T., Ferreyra, C., Giordano, B. L., & Formisano, E. (2026). Multiscale Temporal Processing Supports Sound Recognition under Causal Constraints. *bioRxiv*. <https://doi.org/10.64898/2026.07.27.740874>
- [17] Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). ImageNet classification with deep convolutional neural networks. *Advances in Neural Information Processing Systems*, 25.